
Quantum Physics

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Assignment: Number 3

Problem 1 Evaluate the following commutators

(a) $\left[\sin z, \frac{d}{dz} \right],$

$$\begin{aligned} & \sin z \frac{d}{dz} f - \frac{d}{dz} (\sin z f) \\ &= \sin z \frac{df}{dz} - \cos z f - \sin z \frac{df}{dz} \\ &= -\cos z f \end{aligned}$$

Then $\left[\sin z, \frac{d}{dz} \right] = -\cos z.$

(b) $\left[\frac{d^2}{dx^2}, ax^2 + bx + c \right],$

$$\begin{aligned} & \frac{d^2}{dx^2} \left((ax^2 + bx + c) f \right) - (ax^2 + bx + c) \frac{d^2 f}{dx^2} \\ &= \frac{d}{dx} \left((2ax + b) f + (ax^2 + bx + c) \frac{df}{dx} \right) - (ax^2 + bx + c) \frac{d^2 f}{dx^2} \\ &= 2af + (2ax + b) \frac{df}{dx} + (2ax + b) \frac{df}{dx} + (ax^2 + bx + c) \frac{d^2 f}{dx^2} - (ax^2 + bx + c) \frac{d^2 f}{dx^2} \\ &= \left(2a + 2b \frac{d}{dx} + 4ax \frac{d}{dx} \right) f \end{aligned}$$

Then, $\left[\frac{d^2}{dx^2}, ax^2 + bx + c \right] = 2a + 2b \frac{d}{dx} + 4ax \frac{d}{dx}$

(c) $\left[\frac{d}{dx}, \frac{d^2}{dx^2} \right],$

$$\frac{d}{dx} \frac{d^2 f}{dx^2} - \frac{d^2}{dx^2} \frac{df}{dx} = 0$$

Then, $\left[\frac{d}{dx}, \frac{d^2}{dx^2} \right] = 0$

Problem 2 Classify the following operators as linear or nonlinear

If the operator T is linear, it satisfies:

$$T(au + bv) = aTu + bTv, \quad \text{for all } a, b \in \mathbb{R}, u, v \in \mathcal{U}$$

(a) $3x^2 \frac{d^2}{dx^2}$,

$$\begin{aligned} & 3x^2 \frac{d^2}{dx^2} (au + bv) \\ &= 3x^2 \frac{d^2}{dx^2} (au) + 3x^2 \frac{d^2}{dx^2} (bv) \\ &= a 3x^2 \frac{d^2}{dx^2} (u) + b 3x^2 \frac{d^2}{dx^2} (v) \end{aligned}$$

Then $3x^2 \frac{d^2}{dx^2}$ is a linear operator.

(b) $(\cdot)^2$,

$$(au + bv)^2 = a^2u^2 + abuv + abvu + b^2v^2 \neq (au)^2 + (bv)^2$$

Then $(\cdot)^2$ is not a linear operator.

(c) $\int dx$,

$$\int (au + bv) dx = \int au dx + \int bv dx = a \int u dx + b \int v dx$$

Then $\int dx$ is a linear operator.

(d) $e^{(\cdot)}$,

Consider the special case $a=1, b=e, u=v=1$:

$$e^{au+bv} = e^{1+e} \neq 1 \cdot e + e \cdot e = ae^u + be^v$$

Then $e^{(\cdot)}$ is not a linear operator.

(e) $\frac{\partial}{\partial x} + \frac{\partial}{\partial y}$,

$$\begin{aligned} & \left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} \right) (au + bv) \\ &= \left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} \right) (au) + \left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} \right) (bv) \\ &= a \left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} \right) (u) + b \left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} \right) (v) \end{aligned}$$

Then $\frac{\partial}{\partial x} + \frac{\partial}{\partial y}$ is a linear operator.

(f) $\sum_{x=1}^n (\cdot)$,

$$\sum_{x=1}^n (au(x) + bv(x)) = \sum_{x=1}^n au(x) + \sum_{x=1}^n bv(x) = a \sum_{x=1}^n u(x) + b \sum_{x=1}^n v(x)$$

Then $\sum_{x=1}^n (\cdot)$ is a linear operator.

Problem 3 We define the translation operator $\widehat{T}_h f(x) = f(x+h)$

(a) Check if $\widehat{T}_h$ is a linear operator,

$$\widehat{T}_h (au(x) + bv(x)) = au(x+h) + bv(x+h) = a\widehat{T}_h u(x) + b\widehat{T}_h v(x)$$

Then $\widehat{T}_h$ is a linear operator.

(b) Evaluate $(\widehat{T}_h^2 - 3\widehat{T}_h + 2)x^2$,

$$\begin{aligned} & (\widehat{T}_h^2 - 3\widehat{T}_h + 2)x^2 \\ &= \widehat{T}_h \widehat{T}_h x^2 - 3\widehat{T}_h x^2 + 2x^2 \\ &= \widehat{T}_h (x+h)^2 - 3(x+h)^2 + 2x^2 \\ &= (x+2h)^2 - 3(x+h)^2 + 2x^2 \\ &= x^2 + 4hx + 4h^2 - 3x^2 - 6hx - 3h^2 + 2x^2 \\ &= (-2hx + h^2) \\ &= -2x + 1 \end{aligned}$$

Problem 4 We define the operator $e^{\widehat{A}} = \sum_{k=0}^{\infty} \frac{\widehat{A}^k}{k!}$, show $e^{\widehat{D}} = \widehat{T}_1$, where $\widehat{D} = \frac{d}{dx}$

$$e^{\widehat{D}} f(x) = \sum_{k=0}^{\infty} \frac{\widehat{D}^k f(x)}{k!} = \sum_{k=0}^{\infty} \frac{1}{k!} \left(\frac{d}{dx} \right)^k f(x) = \sum_{k=0}^{\infty} \frac{1}{k!} \frac{d^k f(x)}{dx^k}$$

By Taylor expansion

$$\widehat{T}_1 f(x) = f(x+1) = \sum_{k=0}^{\infty} \frac{1}{k!} \frac{d^k f(x)}{dx^k} \cdot (x+1-x)^k = \sum_{k=0}^{\infty} \frac{1}{k!} \frac{d^k f(x)}{dx^k}$$

Thus $(e^{\widehat{D}} - \widehat{T}_1)f(x) = 0$, for all $f(x)$, hence $e^{\widehat{D}} = \widehat{T}_1$.

Problem 5 Find the eigenfunctions and eigenvalues of $\frac{d^2}{dx^2} + 2\frac{d}{dx}$.

We assume the eigenvalue is λ , and the corresponding eigenfunction is $f(x)$. Then

$$\frac{d^2 f}{dx^2} + 2\frac{df}{dx} = \lambda f.$$

Taking the Laplace transform on both sides, we obtain

$$s^2 F(s) - sf(0) - f'(0) + 2sF(s) - 2f(0) = \lambda F(s),$$

where $f(0)$ and $f'(0)$ are constants depending on the initial or boundary conditions.

Rearranging, we get

$$F(s) = c_1 \frac{s+1}{(s+1)^2 - (\lambda+1)} + c'_2 \frac{1}{(s+1)^2 - (\lambda+1)}.$$

$$F(s) = c_1 \frac{s+1}{(s+1)^2 - (\lambda+1)} + c_2 \frac{\sqrt{\lambda+1}}{(s+1)^2 - (\lambda+1)}.$$

(a) **Case 1:** $1 + \lambda > 0$,

Taking the inverse Laplace transform,

$$f(x) = c_1 e^{-x} \sinh(\sqrt{\lambda+1} x) + c_2 e^{-x} \cosh(\sqrt{\lambda+1} x).$$

(b) **Case 2:** $1 + \lambda < 0$,

$$\sqrt{\lambda+1} = i\sqrt{-(\lambda+1)}.$$

$$\begin{aligned} f(x) &= c_1 e^{-x} \sinh\left(i\sqrt{-(\lambda+1)} x\right) + c_2 e^{-x} \cosh\left(i\sqrt{-(\lambda+1)} x\right) \\ &= A e^{-x} \sin\left(\sqrt{-(\lambda+1)} x\right) + B e^{-x} \cos\left(\sqrt{-(\lambda+1)} x\right). \end{aligned}$$

(c) **Case 3:** $1 + \lambda = 0$,

$$F(s) = c_1 \frac{1}{s+1} + c'_2 \frac{1}{(s+1)^2}.$$

Taking the inverse Laplace transform,

$$f(x) = c_1 e^{-x} + c'_2 x e^{-x}.$$

Problem 6 Evaluate the following commutators

(a) $[\hat{x}, \hat{p}_x]$,

$$x \left(-i\hbar \frac{\partial}{\partial x} \right) f - \left(-i\hbar \frac{\partial}{\partial x} \right) (xf) = -i\hbar x \frac{\partial f}{\partial x} + i\hbar f + i\hbar x \frac{\partial f}{\partial x} = i\hbar f$$

Then $[\hat{x}, \hat{p}_x] = i\hbar$.

(b) $[\hat{x}, \hat{p}_x^2]$,

$$[\hat{x}, \hat{p}_x^2] = [\hat{x}, \hat{p}_x] \hat{p}_x + \hat{p}_x [\hat{x}, \hat{p}_x] = i\hbar \hat{p}_x + \hat{p}_x i\hbar = 2i\hbar \hat{p}_x = 2\hbar^2 \frac{\partial}{\partial x}$$

Then $[\hat{x}, \hat{p}_x^2] = 2i\hbar \hat{p}_x = 2\hbar^2 \frac{\partial}{\partial x}$.

(c) $[\hat{x}, \hat{p}_y],$

$$x \left(-i\hbar \frac{\partial}{\partial y} \right) f - \left(-i\hbar \frac{\partial}{\partial y} \right) (xf) = -i\hbar x \frac{\partial f}{\partial y} + i\hbar x \frac{\partial f}{\partial y} = 0$$

Then $[\hat{x}, \hat{p}_y] = 0.$

(d) $[\hat{x}, \hat{V}(x, y, z)],$

$$xVf - Vxf = 0$$

Then $[\hat{x}, \hat{V}(x, y, z)] = 0.$

(e) $[\hat{x}, \hat{H}],$ where $\hat{H} = \frac{\hat{p}_x^2 + \hat{p}_y^2 + \hat{p}_z^2}{2m} + V(x, y, z)$

$$\begin{aligned} & \left[x, \frac{\hat{p}_x^2 + \hat{p}_y^2 + \hat{p}_z^2}{2m} + V(x, y, z) \right] \\ &= \frac{1}{2m} \left([x, \hat{p}_x^2] + [x, \hat{p}_y^2] + [x, \hat{p}_z^2] \right) + [x, V] \\ &= \frac{1}{2m} \left([\hat{x}, \hat{p}_x^2] + [\hat{x}, \hat{p}_y] \hat{p}_y + \hat{p}_y [\hat{x}, \hat{p}_y] + [\hat{x}, \hat{p}_z] \hat{p}_z + \hat{p}_z [\hat{x}, \hat{p}_z] \right) \\ &= \frac{1}{2m} \left([\hat{x}, \hat{p}_x^2] \right) = \frac{i\hbar \hat{p}_x}{m} = \frac{\hbar^2}{m} \frac{\partial}{\partial x} \end{aligned}$$

Then $[\hat{x}, \hat{H}] = \frac{i\hbar \hat{p}_x}{m} = \frac{\hbar^2}{m} \frac{\partial}{\partial x}.$

(f) $[\hat{x}\hat{y}\hat{z}, \hat{p}_x^2],$

$$\begin{aligned} & [\hat{x}\hat{y}\hat{z}, \hat{p}_x^2] \\ &= -\hbar^2 xyz \frac{\partial^2 f}{\partial x^2} + \hbar^2 \frac{\partial^2}{\partial x^2} (xyzf) \\ &= -\hbar^2 xyz \frac{\partial^2 f}{\partial x^2} + \hbar^2 \frac{\partial}{\partial x} \left(yzf + xyz \frac{\partial f}{\partial x} \right) \\ &= -\hbar^2 xyz \frac{\partial^2 f}{\partial x^2} + \hbar^2 \left(yz \frac{\partial f}{\partial x} + yz \frac{\partial f}{\partial x} + xyz \frac{\partial^2 f}{\partial x^2} \right) \\ &= 2\hbar^2 yz \frac{\partial}{\partial x} f = 2i\hbar yz \hat{p}_x f \end{aligned}$$

Then $[\hat{x}\hat{y}\hat{z}, \hat{p}_x^2] = 2i\hbar yz \hat{p}_x = 2\hbar^2 yz \frac{\partial}{\partial x}.$